Notes on Chapter 7

Throughout, $m > 1$ is the modulus and a is an integer. The subject of these notes is the set of invertible elements modulo m, which is denoted U_m. The main result, Theorem 23, gives a complete classification of those m for which there exists an element $\zeta \in U_m$ such that U_m is equal to $\{\zeta^i : 1 \leq i \leq \phi(m)\}$. In case this is true, we say ζ is a primitive root modulo m. This is an important component of the complete description of the group U_m. For a complete description of the groups U_m, see [2, Remark 2.3.31].

Definition 1. ([1, Definition 7-1]) If n is the least positive integer such that $a^n \equiv 1 \pmod{m}$, then we say the *order of a modulo m* is n. We write $|a| = n$.

Theorem 2. *(*[1, Theorem 7-1]*) The order of a modulo m exists if and only if* $\gcd(a, m) = 1$.

Proof. If $\gcd(a, m) = 1$, then $a^{\phi(m)} \equiv 1 \pmod{m}$, by Euler's Theorem. Since $\phi(m) \geq 1$, the set $\{x \geq 1 : a^x \equiv 1 \pmod{m}\}$ is nonempty. By the Well Ordering Principle, the order of a exists. Conversely, if $a^h \equiv 1 \pmod{m}$, then

$$aa^{h-1} = 1 + mq$$

for some q. By Bézout's Identity, $\gcd(a, m) = 1$. □

Theorem 3. *(*[1, Theorem 7-2]*) Suppose the order of a modulo m is n and t is an integer. Then $a^t \equiv 1 \pmod{m}$ if and only if $n \mid t$.*

Proof. Suppose $a^t \equiv 1 \pmod{m}$. Apply the Division Algorithm to write $t = hq + r$, where $0 \leq r < n$. Then

$$1 \equiv a^t \equiv (a^n)^q a^r \equiv a^r \pmod{m}.$$

By the minimal choice of n, we conclude that $r = 0$. Conversely, if $t = nq$, then $a^t = (a^n)^q \equiv 1 \pmod{m}$. □

Corollary 4. *(Corollary to* [1, Theorem 7-2]*) If the order of a modulo m is n, then $n \mid \phi(m)$.*

Proof. This is immediate, by Euler's Theorem and Theorem 3. □

Corollary 5. *(Corollary to* [1, Theorem 7-2]*) If the order of a modulo m is n, then $a^i \equiv a^j \pmod{m}$ if and only if $i \equiv j \pmod{n}$.*

Proof. If $i \equiv j \pmod{n}$, then $i = j + nq$ for some q. Then

$$a^i = (a^n)^q a^j \equiv a^j \pmod{m}.$$

If $a^i \equiv a^j \pmod{m}$, then $a^{i-j} \equiv 1 \pmod{m}$. By Theorem 3, $n \mid (i - j)$. Hence $i \equiv j \pmod{n}$. □

Definition 6. If the order of ζ modulo m is $\phi(m)$, then we say ζ is a *primitive root modulo m*.

Theorem 7. *(*[1, Theorem 7-3]*) If ζ is a primitive root modulo m, then the set*

$$R = \left\{\zeta, \zeta^2, \ldots, \zeta^{\phi(m)}\right\}$$

is a reduced residue system modulo m.

Proof. By Corollary 5, elements of R are pairwise incongruent. Hence the set R has $\phi(m)$ elements. Define $f : R \to U_m$ by setting $f(\zeta^i)$ equal to the remainder upon division of ζ^i by m. Then f is one-to-one by Corollary 5. Both sets have cardinal number $\phi(m)$, so the Pigeon Hole Principle says f is onto. □

Theorem 8. *(*[1, Theorem 7-4]*) If the order of a modulo m is n, then the order of a^k is equal to $n/\gcd(k,n)$.*

Proof. Set $|a^k| = \ell$. Then $a^{k\ell} \equiv 1 \pmod{m}$. Theorem 3 says $n \mid k\ell$. Write $k\ell = nx$ for some x. Let $d = \gcd(k,n)$. Factor: $k = dk_1$ and $n = dn_1$. We prove that $\ell = n_1$. By Bézout's Identity, $d = ku + nv$ for some u, v. Cancel d from both sides to get $1 = k_1u + n_1v$. Therefore, $\gcd(k_1, n_1) = 1$. Cancel d from $k\ell = nx$ to get $k_1\ell = n_1x$. By Euclid's Lemma, $n_1 \mid \ell$. Now look at

$$\left(a^k\right)^{n_1} \equiv a^{dk_1n_1} \equiv (a^n)^{k_1} \equiv 1 \pmod{m}.$$

Theorem 3 says ℓ divides n_1. Taken together, this shows $\ell = n_1$. □

Corollary 9. *(*[1, Corollary 7-1]*) If ζ is a primitive root modulo m, then ζ^i is a primitive root modulo m if and only if $\gcd(i, \phi(m)) = 1$.*

Proof. The order of ζ^i is equal to $\phi(m)$ if and only if $\gcd(i, \phi(m)) = 1$. □

Corollary 10. *(Corollary to* [1, Theorem 7-4]*) Suppose the order of a modulo m is n. The set $\left\{|a^i| : 1 \le i \le n\right\}$ is equal to the set* $\mathrm{Div}(n)$.

Proof. By Theorem 8, the order of a^i is equal to $n/\gcd(i,n)$, which is a divisor of n. If $d \mid n$, then by Theorem 8, the order of $a^{n/d}$ is equal to d. □

Definition 11. If the order of a modulo m is n, define the set

$$\langle a \rangle = \{a, a^2, \ldots, a^n\}.$$

Corollary 12. *(Corollary to* [1, Theorem 7-4]*) Suppose the order of a modulo m is n. For any d in* $\mathrm{Div}(n)$*, there are exactly $\phi(d)$ elements of order d in $\langle a \rangle$.*

Proof. In the notation of Theorem 6-1, set

$$T_{n/d} = \{k : 1 \le k \le n \text{ and } \gcd(k,n) = n/d\}.$$

The elements of order d in $\langle a \rangle$ is the set

$$\left\{a^k : 1 \le k \le n \text{ and } |a^k| = d\right\},$$

which by Theorem 8 is equal to the set

$$\left\{a^k : 1 \le k \le n \text{ and } \gcd(k,n) = n/d\right\},$$

which is equal to

$$\left\{a^k : 1 \le k \le n \text{ and } k \in T_{n/d}\right\}.$$

The cardinality of the last set is $\#\left(T_{n/d}\right)$, which by Theorem 6-1 is equal to $\phi(d)$. □

Theorem 13. *(*[1, Theorem 7-5]*) If there exists a primitive root modulo m, then there are exactly $\phi(\phi(m))$ mutually incongruent primitive roots modulo m.*

Proof. Apply Corollary 12 with $d = n = \phi(m)$. □

Corollary 14. *(Corollary to* [1, Theorem 7-4]*) If a_1 and a_2 are invertible modulo m, the order of a_1 modulo m is d_1, the order of a_2 modulo m is d_2, and $\gcd(d_1, d_2) = 1$, then the following are true.*

(1) $\langle a_1 \rangle \cap \langle a_2 \rangle = \{1\}$. That is, $a_1^i \equiv a_2^j \pmod{m}$ if and only if $a_1^i \equiv 1 \pmod{m}$ and $a_2^j \equiv 1 \pmod{m}$.
(2) The order of $a_1 a_2$ modulo m is $d_1 d_2$.

Proof. (1): Suppose $a_1^i \equiv a_2^j \pmod{m}$ for some i and j. Let $d = |a_1^i| = |a_1^i|$. Theorem 8 says $d \mid d_1$ and $d \mid d_2$. Therefore, $d \leq \gcd(d_1, d_2) = 1$. Hence $a_1^i \equiv 1 \pmod{m}$ and $a_2^j \equiv 1 \pmod{m}$. The converse is immediate by [1, Theorem 4-1].

(2): By Exercise 4 of §2.2, $\operatorname{lcm}(d_1, d_2) = d_1 d_2$. Let $|a_1 a_2| = d$. Then

$$(a_1 a_2)^{d_1 d_2} = \left(a_1^{d_1}\right)^{d_2} \left(a_2^{d_2}\right)^{d_1} \equiv 1 \pmod{m}.$$

Theorem 3 says $d \mid d_1 d_2$. Since $(a_1 a_2)^d \equiv 1 \pmod{m}$, we have $a_1^d \equiv a_2^{-d} \pmod{m}$. Part (1) together with Theorem 3 imply that $d_1 \mid d$ and $d_2 \mid d$. Hence, $d_1 d_2 = \operatorname{lcm}(d_1, d_2) \leq d$. Taken together we have $d_1 d_2 = d$. □

Corollary 15. *(Corollary to* [1, Theorem 7-4]*) If $m > 2$ and $n > 2$ and $\gcd(m, n) = 1$, then there is no primitive root modulo mn.*

Proof. By Exercise 4 of § 6-1, $\phi(m)$ is even and $\phi(n)$ is even. Because $m > 2$, $1 \not\equiv -1 \pmod{m}$. There are at least two solutions modulo m to $x^2 \equiv 1 \pmod{m}$, namely $x \equiv \pm 1 \pmod{m}$. Likewise, there are at least two solutions modulo n to $x^2 \equiv 1 \pmod{n}$, namely $x \equiv \pm 1 \pmod{n}$. By the Chinese Remainder Theorem, there is a unique x_1 in U_{mn} which satisfies

$$\begin{cases} x_1 \equiv -1 \pmod{m} \\ x_1 \equiv 1 \pmod{n} \end{cases}$$

Likewise, there exist unique x_2, x_3, x_4 in U_{mn} which satisfy

$$\begin{cases} x_2 \equiv 1 \pmod{m} \\ x_2 \equiv -1 \pmod{n} \end{cases}, \begin{cases} x_3 \equiv -1 \pmod{m} \\ x_3 \equiv -1 \pmod{n} \end{cases}, \begin{cases} x_3 \equiv 1 \pmod{m} \\ x_3 \equiv 1 \pmod{n} \end{cases}.$$

Hence x_1, x_2, x_3, x_4 are four mutually incongruent solutions to $x^2 \equiv 1 \pmod{mn}$. For contradiction's sake, assume ζ is a primitive root modulo mn. By Corollary 12, there are only two solutions to $x^2 \equiv 1 \pmod{mn}$, namely $x \equiv \pm 1 \pmod{mn}$. This is a contradiction, which implies a primitive root does not exist. □

Theorem 16. *For all $k \geq 3$ the following are true.*

(1) For any odd integer a, $a^{2^{k-2}} \equiv 1 \pmod{2^k}$.
(2) The order of a modulo 2^k is less than or equal to $a^{2^{k-2}}$.
(3) There is no primitive root modulo 2^k.

Proof. (1): The proof is by induction on k. For $k = 3$, this follows from direct computation: $1^2 \equiv 3^2 \equiv 5^2 \equiv 7^2 \equiv 1 \pmod{8}$. Inductively assume $k \geq 3$, a is odd, and $a^{2^{k-2}} \equiv 1 \pmod{2^k}$. Then $a^{2^{k-2}} = 1 + x2^k$ for some integer x. Squaring both

sides leads to the following.

$$\begin{aligned} a^{2^{k-1}} &= 1 + 2x2^k + x^2 2^{2k} \\ &= 1 + \left(x + x^2 2^{k-1}\right) 2^{k+1} \\ &\equiv 1 \pmod{2^{k+1}}. \end{aligned}$$

This and Mathematical Induction prove (1).

(2): This is straight from Part (1) and Definition 1.

(3): The set of units modulo 2^k is equal to $U_{2^k} = \{u : 1 \leq u \leq 2^k \text{ and } u \text{ is odd}\}$. By [1, Example 6-1], $\phi(2^k) = 2^{k-1}$. By (2) the order of every unit in U_{2^k} is less than $\phi(2^k)$. □

Theorem 17. *(*[1, Theorem 7-6]*) If p is a prime, then there exists a primitive root modulo p.*

Proof. If $p = 2$, then $\phi(2) = 1$ and 1 is a primitive root. From now on assume p is an odd prime. Let $u \in U_p$ and assume the order of u is n modulo p. By Corollary 5, the set $\langle u \rangle = \{u, u^2, \dots, u^n\}$ is a set of n numbers pairwise incongruent modulo p. For any integer i, the order of u^i divides n, by Theorem 8. By Theorem 3, every u^i is a solution to the polynomial congruence

$$x^n \equiv 1 \pmod{p}. \tag{1}$$

So $\{u, u^2, \dots, u^n\}$ is a set of n mutually incongruent solutions to Eq. (1). By [1, Theorem 5-5], the polynomial congruence Eq. (1) has no more than n such solutions. Thus every solution to Eq. (1) is congruent to an element of $\{u, u^2, \dots, u^n\}$. In particular, every element of order n in U_p is congruent to an element of $\{u, u^2, \dots, u^n\}$. In Corollary 12 we proved that the number of elements in $\{u, u^2, \dots, u^n\}$ of order n is equal to $\phi(n)$. Define an arithmetic function $N : \{1, 2, 3, \dots\} \to \mathbb{N}$ by setting $N(n) = \#\{a \in U_p : |a| = n\}$. Thus $N(n)$ is the number of elements in U_p of order n. We have shown that if $N(n) \neq 0$, then $N(n) = \phi(n)$. By Corollary 4, if $a \in U_p$, then $|a|$, the order of a, is a divisor of $\phi(p) = p - 1$. In particular, we have $N(n) \leq \phi(n)$, for all $n \in \mathrm{Div}(p-1)$. We have

$$p - 1 = \sum_{n|(p-1)} N(n) \leq \sum_{n|(p-1)} \phi(n) = p - 1$$

where the right-most equation is by [1, Theorem 6-1]. Subtracting

$$0 = \sum_{n|(p-1)} (\phi(n) - N(n))$$

where the right hand side is a sum of non-negative integers. We conclude that for every $n \in \mathrm{Div}(p-1)$ there are exactly $N(n) = \phi(n)$ elements of order n in U_p. In particular, there are $\phi(p-1)$ primitive roots modulo p. □

Lemma 18. *If p is an odd prime, then there exists a primitive root ζ modulo p such that $\zeta^{p-1} \not\equiv 1 \pmod{p^2}$.*

Proof. Let ζ be an integer such that the order of ζ is $p - 1$ modulo p. Assume $\zeta^{p-1} \equiv 1 \pmod{p^2}$. Note that $\zeta + p \equiv \zeta \pmod{p^2}$, hence $\zeta + p$ is another primitive root modulo p. We show that $\zeta + p$ satisfies the conclusion of the lemma. From the Binomial Theorem we have

$$(\zeta + p)^{p-1} = \zeta^{p-1} + (p-1)\zeta^{p-2}p + \text{terms with a factor of } p^2.$$

Reducing modulo p^2, we get

$$\begin{aligned}(\zeta+p)^{p-1} &\equiv \zeta^{p-1}+(p-1)\zeta^{p-2}p \pmod{p^2}\\ &\equiv 1+\zeta^{p-2}p^2-\zeta^{p-2}p \pmod{p^2}\\ &\equiv 1-\zeta^{p-2}p \pmod{p^2}.\end{aligned}$$

Now $(\zeta+p)^{p-1}\not\equiv 1 \pmod{p^2}$ because $\gcd(\zeta,p)=1$. □

Lemma 19. *If $0\le k\le n$, then $\binom{n}{k}$ is a positive integer.*

Proof. This is a proof by Mathematical Induction on the variable n. We proved that $\binom{n}{k}=1$ if $k=n$ or $k=0$. In other words, on the boundaries of Pascal's Triangle, the numbers are all equal to 1. This serves as a basis for the induction. Inductively assume $n\ge 1$ and that the numbers in row n of Pascal's triangle are positive integers. In other words, assume $\binom{n}{i}$ is a positive integer for all i such that $0\le i\le n$. By Pascal's Identity, for any k such that $1\le k\le n$, $\binom{n+1}{k}=\binom{n}{k-1}+\binom{n}{k}$, which is a positive integer. Therefore, every number in row $n+1$ of Pascal's Triangle is a positive integer. □

Lemma 20. *If p is prime and $1\le k\le p-1$, then $\binom{p}{k}\equiv 0 \pmod{p}$.*

Proof. By definition, $\binom{p}{k}=\frac{p!}{k!(p-k)!}=\frac{p(p-1)!}{k!(p-k)!}$. We know from Pascal's Identity that $\binom{p}{k}$ is an integer. That is, the denominator divides the numerator. The denominator, $k!(p-k)!$, is relatively prime to p, since p is prime. By Euclid's Lemma, the denominator divides $(p-1)!$. In other words, $\frac{(p-1)!}{k!(p-k)!}$ is an integer. □

Lemma 21. *If p is an odd prime and ζ is a primitive root modulo p such that $\zeta^{p-1}\not\equiv 1 \pmod{p^2}$, then for each $k\ge 2$,*

$$\left(\zeta^{p-1}\right)^{p^{k-2}}\not\equiv 1 \pmod{p^k}.$$

Proof. The proof is by Mathematical Induction, on the variable k. The basis step, $k=2$, is true by hypothesis. Fix $k\ge 2$ and assume $\left(\zeta^{p-1}\right)^{p^{k-2}}\not\equiv 1 \pmod{p^k}$. Note that $1=\gcd(\zeta,p)=\cdots=\gcd(\zeta,p^{k-1})=\gcd(\zeta,p^k)$. By [1, Example 6-1], we have $\phi(p^{k-1})=p^{k-2}(p-1)$, and Euler's Theorem ([1, Theorem 5-2]) says

$$\left(\zeta^{p-1}\right)^{p^{k-2}}\equiv 1 \pmod{p^{k-1}}.$$

For some integer x we have

$$\left(\zeta^{p-1}\right)^{p^{k-2}}=1+xp^{k-1}.$$

By our induction hypothesis we know p does not divide x. Raise both sides to the power p. Expand the right hand side using the Binomial Theorem to get

$$\begin{aligned}\left(\zeta^{p-1}\right)^{p^{k-1}} &= \left(1+xp^{k-1}\right)^p\\ &= 1+pxp^{k-1}+\sum_{i=2}^{p-1}\binom{p}{i}x^ip^{i(k-1)}+x^pp^{p(k-1)}.\end{aligned}\tag{2}$$

Since $k-2\ge 0$, we see that $k+k-2\ge k$, or $2(k-1)\ge k$. More generally, for any $i\ge 2$, it is true that $i(k-1)\ge k$. Therefore, for $2\le i<p$

$$\binom{p}{i}x^ip^{i(k-1)}=\frac{p(p-1)!}{i!(p-i)!}p^{i(k-1)}x^i=\frac{(p-1)!}{i!(p-i)!}p^{i(k-1)+1}x^i$$

is a multiple of p^{k+1}. For $i = p$, it follows from $2(k-1) \geq k$ that $p(k-1) \geq (p/2)k$. But $p/2 \geq 3/2$, so $p(k-1) \geq (3/2)k = k + k/2$. But $k \geq 2$, so $p(k-1) \geq k+1$. This shows the term $p^{p(k-1)}x^p$ is a multiple of p^{k+1}. Reducing Eq. (2) modulo p^{k+1}, we have

$$\left(\zeta^{p-1}\right)^{p^{k-1}} \equiv 1 + xp^k \pmod{p^{k+1}}.$$

Since p does not divide x, this proves $\left(\zeta^{p-1}\right)^{p^{k-1}} \not\equiv 1 \pmod{p^{k+1}}$. □

Theorem 22. *If p is an odd prime and $a \geq 1$, then the following are true.*

(1) There exists a primitive root modulo p^a.
(2) There exists a primitive root modulo $2p^a$.

Proof. (1): By Lemmas 18 and 21, there exists a primitive root ζ for p such that $\left(\zeta^{p-1}\right)^{p^{k-2}} \not\equiv 1 \pmod{p^k}$. Denote the order of ζ modulo p^k by n. Then n divides $\phi(p^k) = p^{k-1}(p-1)$. Since $\zeta^n \equiv 1 \pmod{p^k}$, this implies $\zeta^n \equiv 1 \pmod{p}$. But ζ is a primitive root for p, so $p-1$ divides n. Write $n = (p-1)p^\ell$ for some ℓ such that $0 \leq \ell \leq k-1$. For sake of contradiction assume $\ell \leq k-2$. Then n divides $p^{k-2}(p-1)$. By Theorem 3

$$\left(\zeta^{p-1}\right)^{p^{k-2}} = \zeta^{p^{k-2}(p-1)} \equiv 1 \pmod{p^k}.$$

This contradiction implies $\ell = k-1$ and ζ is a primitive root for p^k.

(2): By [1, Theorem 6-4], ϕ is multiplicative. Hence $\phi(2p^a) = \phi(2)\phi(p^a) = \phi(p^a)$. Let ζ be a primitive root for p^a, which exists by Part (1). By the Chinese Remainder Theorem there exists r in U_{2p^a} such that

$$\begin{cases} r & \equiv \zeta \pmod{p^a} \\ r & \equiv 1 \pmod{2}. \end{cases}$$

Let d denote the order of r modulo $2p^a$. By Corollary 4, d divides $\phi(2p^a) = \phi(p^a)$. Since $r^d \equiv 1 \pmod{2p^a}$, it follows that $r^d \equiv 1 \pmod{p^a}$. Since $r \equiv \zeta \pmod{p^a}$ and the order of ζ is $\phi(p^a)$, Theorem 3 implies $\phi(p^a)$ divides d. Taken together, this proves $d = \phi(p^a) = \phi(2p^a)$. □

Theorem 23. *A primitive root exists modulo m if and only if m is in one of the following forms: $m = 2$, $m = 4$, $m = p^a$, or $m = 2p^a$, where p is an odd prime and $a \geq 1$.*

Proof. A primitive root for $m = 2$ is 1. A primitive root for $m = 4$ is -1. If $m = p^a$ or $m = 2p^a$, then by Theorem 22 a primitive root exists modulo m. Now suppose m is not one of the forms listed. If m is a prime power, then $m = 2^k$, where $k \geq 3$. By Theorem 16, there is no primitive root modulo 2^k. Suppose there are two distinct odd primes, say p and q, that divide m. Then $m = p^e x$, where $q \mid x$ and $\gcd(p, x) = 1$. By Corollary 15, there is no primitive root modulo m. The last case is $m = 2^k x$ where $k > 1$, x is odd, and $x > 1$. Again, by Corollary 15, there is no primitive root modulo m. □

References

[1] George E. Andrews, *Number theory*, Dover Publications, Inc., New York, 1994, Corrected reprint of the 1971 original [Dover, New York; MR0309838 (46 #8943)]. MR 1298627

[2] Timothy J. Ford, *Introduction to abstract algebra*, Open Math Notes, American Mathematical Society, Reference # OMN:202006.110827, 2024, available at the URL: `https://www.ams.org/open-math-notes/omn-view-listing?listingId=110827`.